CYTECH ACADEMY

Network Security

Complete Study Notes - beginner friendly, exam and lab ready.

(c) CyTech. For enrolled students. Unauthorized attack is illegal - practice only in labs.

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Networking Essentials

Master the OSI/TCP-IP model: how a packet travels application -> transport (TCP/UDP, ports) -> network (IP, routing) -> link (MAC, switches). Know the TCP 3-way handshake (SYN, SYN-ACK, ACK) - scanning and firewall evasion both build on it.

Key ports: 21 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 80/443 HTTP(S), 110/143/993 mail, 139/445 SMB, 3306 MySQL, 3389 RDP, 5432 Postgres, 8080 proxy.

Packet Analysis with Wireshark

Capture, then filter: ip.addr==10.0.0.5, tcp.port==80, dns, http.request, tcp.flags.syn==1. Follow TCP Stream reconstructs a whole conversation - credentials in plaintext protocols (FTP/Telnet/HTTP) appear instantly.

Learn to spot: ARP spoofing storms, DNS tunneling (huge TXT queries), beaconing (regular identical intervals = C2), port scans (many SYNs, no ACK).

Firewalls, IDS/IPS, Segmentation

Firewalls: stateless vs stateful vs next-gen. Default-deny inbound, allow only what the business needs, log everything denied. Segment networks (VLANs, DMZ) so one compromised host cannot reach the crown jewels.

IDS (Snort/Suricata) alerts; IPS blocks. Tune signatures to your environment or you drown in false positives.

VPNs, Proxies and DNS

A VPN encrypts all traffic between you and the VPN server - your ISP sees only ciphertext to one destination. Protocols: WireGuard (fast, modern), OpenVPN (mature), IPsec (enterprise). A VPN protects transport privacy; it does NOT make you anonymous by itself.

A proxy forwards requests for specific apps (browser). SOCKS5 (e.g. via ssh -D 1080) is flexible; HTTPS intercepting proxies (Burp, ZAP) let you inspect and modify your own web traffic - the core pentester workflow. Tor routes via 3 volunteer nodes for anonymity.

DNS: use encrypted DNS (DoH/DoT) or a resolver you trust; block malicious domains at DNS level (Pi-hole). Attackers abuse DNS for exfiltration, so log and alert on unusual query volume/length.

Hardening a Network

Kill legacy protocols (Telnet, FTP, HTTP, SMBv1), enforce WPA3/WPA2-Enterprise, 802.1X port auth, strong SNMPv3, patched firmware, central logging to a SIEM (Wazuh/Splunk free tier), and an incident response plan tested with tabletop exercises.